Exp 8 MATLAB CODE

```
clc;
clear;
close all;

Fs = 100000;
t = 0:1/Fs:0.01;

Am = 1;
fm = 1000;
fc = 10000;
kp = pi/2;

% Message Signal
m = Am*cos(2*pi*fm*t);

% Phase Modulated Signal
pm = cos(2*pi*fc*t + kp*m);

% Power Spectral Density using FFT
N = length(pm);
Y = fft(pm);
P = abs(Y/N).^2;
P = P(1:floor(N/2)+1);
P(2:end-1) = 2*P(2:end-1);
f = Fs*(0:floor(N/2))/N;

% Plot Message Signal
```

```
figure;

subplot(3,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Plot PM Signal
subplot(3,1,2);
plot(t,pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Plot PSD
subplot(3,1,3);
plot(f,10*log10(P + eps));
title('Power Spectral Density of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Power (dB)');
xlim([0 20000]);
grid on;

% Display Parameters
fprintf('Carrier Frequency = %d Hz\n',fc);
```

```
fprintf('Message Frequency = %d Hz\n',fm);

fprintf('Phase Deviation = %.2f radians\n',kp*Am);

fprintf('Phase Modulation Index = %.2f\n',kp*Am);
```

